

WEEK-7

AIM

To design and implement a full stack web application named **StudentHub** that enables a student to register, have their details stored persistently in a JSON file, log in using their username or email along with a password, view their personal and academic information on a dashboard, and log out — thereby demonstrating the complete frontend-to-backend data flow using Node.js, Express.js, and a JSON file as the data store.

DESCRIPTION

StudentHub is a student management web application built using a full stack approach with plain HTML, CSS, and JavaScript on the frontend, and Node.js with the Express.js framework on the backend. Instead of a conventional database such as MySQL or MongoDB, the application uses a JSON file (users.json) as a lightweight, file-based data store, making the project simple enough to understand and explain during a viva while still covering every layer of a real web application.

The frontend consists of four pages — a landing page, a registration page, a login page, and a dashboard — all styled with a single shared stylesheet. The registration page collects a student's personal, contact, academic, and account details through an HTML form, validates them in the browser using JavaScript, and sends the data to the server using the Fetch API. The Express server receives this data through req.body, performs its own validation (including checks for duplicate usernames and emails), and appends the new student record to the array stored in users.json without disturbing previously registered students.

The login page accepts either a username or an email along with a password. The server reads users.json, searches for a matching record, and verifies the password before returning the student's details (excluding the password) to the browser. On a successful login, this data is saved in the browser's localStorage, which the dashboard page reads on load to display the student's profile — personal name, email, username, mobile number, date of birth, gender, course, year of study, and address. If no logged-in user is found in localStorage, the dashboard automatically redirects to the login page, preventing unauthorized access. A logout button clears the stored session data and returns the user to the login page.

Through this flow, the project demonstrates core full stack concepts: HTML forms and DOM manipulation, client-side validation, asynchronous communication via fetch(), Express routing and middleware, handling of POST request bodies, Node.js file-system operations for reading and writing JSON, and the use of browser localStorage to maintain a simple client-side session — all without relying on frameworks, databases, or authentication libraries beyond what the experiment requires.

PROGRAM CODE

index.html:

```
<!DOCTYPE html>

<html lang="en">

<head>

  <meta charset="UTF-8" />

  <meta name="viewport" content="width=device-width, initial-scale=1.0" />

  <title>StudentHub — Manage Your Student Journey</title>

  <link rel="stylesheet" href="css/style.css" />

</head>

<body>

  <nav class="navbar">

    <div class="container">

      <a href="/" class="brand">

        <span class="brand-mark">SH</span>

        <span class="brand-text">

          <span class="brand-name">StudentHub</span>

          <span class="brand-tagline">Smart • Simple • Student-Centric</span>

        </span>

      </a>

      <div class="nav-links">

        <a href="/index.html">Home</a>

        <a href="/register.html">Register</a>

        <a href="/login.html">Login</a>

      </div>

      <div class="nav-actions">

        <a href="/login.html" class="btn btn-outline">Login</a>

        <a href="/register.html" class="btn btn-primary">Get Started</a>

      </div>

    </div>

  </nav>

</body>

</html>
```

</div>

</nav>

<header class="hero">

<div class="container">

<div class="hero-copy">

<h1>Manage Your Student Journey</h1>

<p class="lead">

StudentHub keeps your academic profile in one place — register once,
log in anytime, and view your personal and academic information from
a clean, simple dashboard.

</p>

<div class="hero-actions">

Get Started

Login

</div>

</div>

<div class="id-card-wrap">

<div class="id-card" aria-hidden="true">

<div class="id-card-top">

Student ID

SH

</div>

<div class="id-card-photo">R</div>

<div class="id-card-body">

<h3>Rahul Kumar</h3>

<p>B.Tech CSE · 3rd Year</p>

</div>

<div class="id-card-footer">

ID 2026–0148

</div>

```
</div>

</div>

</div>

</header>

<section class="steps">
  <div class="container">
    <div class="steps-intro">
      <h2>Three steps, start to finish</h2>
      <p>From first registration to your personal dashboard, the whole journey happens in one place.</p>
    </div>
    <ol class="steps-list">
      <li class="step-item">
        <span class="step-index">1</span>
        <h3>Easy Registration</h3>
        <p>Quickly create a student profile with your personal, contact, and academic details.</p>
      </li>
      <li class="step-item">
        <span class="step-index">2</span>
        <h3>Secure Login</h3>
        <p>Access the student dashboard using your username or email along with your password.</p>
      </li>
      <li class="step-item">
        <span class="step-index">3</span>
        <h3>Student Dashboard</h3>
        <p>View your personal and academic information in one clean, organized profile card.</p>
      </li>
    </ol>
  </div>
</section>

<footer class="site-footer">
  <div class="container">
```

StudentHub · & Academic Full Stack Development project.

Built with HTML, CSS, JavaScript, Node.js & Express.js

</body>

dashboard.html

```
<!DOCTYPE html>
```

```
<html lang="en">
```

<head>

<meta charset="UTF-8" />

```
<meta name="viewport" content="width=device-width, initial-scale=1.0" />
```

<title>Dashboard — StudentHub</title>

```
<link rel="stylesheet" href="css/style.css" />
```

</head>

<body>

```
<nav class="navbar">
```

<div class="container">

[](/)

SH

StudentHub

Smart • Simple • Student-Centric

<div class="nav-links">

`<a href="/">Home</a>`

<div class="nav-actions">

<button id="logoutBtn" class="btn btn-danger">Logout</button>

</div>

</div>

</nav>

<main class="dashboard-page">

<div class="container">

<div id="loadingState" class="dashboard-loading">

<p>Loading your dashboard...</p>

</div>

<div id="dashboardContent" style="display:none;">

<div class="dashboard-header">

<div>

<h1 id="welcomeHeading">Welcome!</h1>

<p>Here is your student profile on record.</p>

</div>

</div>

<div class="profile-card">

<div class="profile-card-head">

<div class="profile-avatar" id="avatarInitial">S</div>

<div>

<h2 id="profileName">Student Name</h2>

Course • Year

</div>

</div>

<div class="profile-list">

<div class="profile-row">

Full Name

—

</div>

<div class="profile-row">

Email

—

</div>

<div class="profile-row">

Username

—

</div>

<div class="profile-row">

Mobile

—

</div>

<div class="profile-row">

Date of Birth

—

</div>

<div class="profile-row">

Gender

—

</div>

<div class="profile-row">

Course

—

</div>

<div class="profile-row">

Year of Study

—

</div>

<div class="profile-row">

Address

—

</div>

</div>

</div>

```
</div>
</div>
</main>
<script src="js/script.js"></script>
</body>
</html>
```

login.html:

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8" />
  <meta name="viewport" content="width=device-width, initial-scale=1.0" />
  <title>Login — StudentHub</title>
  <link rel="stylesheet" href="css/style.css" />
</head>
<body>
  <nav class="navbar">
    <div class="container">
      <a href="/" class="brand">
        <span class="brand-mark">SH</span>
        <span class="brand-text">
          <span class="brand-name">StudentHub</span>
          <span class="brand-tagline">Smart • Simple • Student-Centric</span>
        </span>
      </a>
      <div class="nav-links">
        <a href="index.html">Home</a>
        <a href="./register.html">Register</a>
        <a href="login.html">Login</a>
      </div>
    </div>
  </nav>

```

```
</div>

<div class="nav-actions">

  <a href="register.html" class="btn btn-primary">Get Started</a>

</div>

</div>

</nav>

<main class="auth-split">

  <aside class="auth-side">

    <div class="auth-side-inner">

      <div class="side-mark">SH</div>

      <h2>Welcome back to your record</h2>

      <p>Log in to pick up right where you left off — your profile, course, and details, exactly as you
registered them.</p>

      <blockquote>&ldquo;One login, your whole student profile.&rdquo;</blockquote>

    </div>

  </aside>

  <div class="auth-form-panel">

    <div class="auth-form-panel-inner">

      <div class="auth-head">

        <h1>Login</h1>

        <p>Enter your credentials to access your dashboard.</p>

      </div>

      <div id="formMessage" class="message-box"></div>

      <form id="loginForm" novalidate>

        <div class="field">

          <label for="identifier">Username or Email</label>

          <input type="text" id="identifier" name="identifier" placeholder="e.g. rahul123 or
rahul@gmail.com" />

          <span class="field-error-text" data-error-for="identifier"></span>

        </div>

        <div class="field">

          <label for="password">Password</label>
```

```
<input type="password" id="password" name="password" placeholder="Your password" />
<span class="field-error-text" data-error-for="password"></span>
</div>
<button type="submit" class="btn btn-primary btn-block" id="loginBtn">Login</button>
<div class="form-footer-link">
  Don't have an account? <a href="./register.html">Register</a>
</div>
</form>
</div>
</div>
</main>
<script src="js/script.js"></script>
</body>
</html>
```

register.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8" />
  <meta name="viewport" content="width=device-width, initial-scale=1.0" />
  <title>Register — StudentHub</title>
  <link rel="stylesheet" href="css/style.css" />
</head>
<body>
  <nav class="navbar">
    <div class="container">
      <a href="/" class="brand">
        <span class="brand-mark">SH</span>
        <span class="brand-text">
          <span class="brand-name">StudentHub</span>
          <span class="brand-tagline">Smart • Simple • Student-Centric</span>
        </span>
      </a>
    </div>
  </nav>
</body>
```

```
</span>
</a>
<div class="nav-links">
  <a href="index.html">Home</a>
  <a href="register.html">Register</a>
  <a href="login.html">Login</a>
</div>
<div class="nav-actions">
  <a href="login.html" class="btn btn-outline">Login</a>
</div>
</div>
</nav>
<main class="auth-page">
  <div class="auth-card">
    <div class="auth-head">
      <h1>Create your student account</h1>
      <p>Fill in your details below to register with StudentHub.</p>
    </div>
    <div id="formMessage" class="message-box"></div>
    <form id="registerForm" novalidate>
      <div class="form-section-title">Personal Information</div>
      <div class="form-grid">
        <div class="field full">
          <label for="name">Full Name</label>
          <input type="text" id="name" name="name" placeholder="e.g. Rahul Kumar" />
          <span class="field-error-text" data-error-for="name"></span>
        </div>
        <div class="field">
          <label for="age">Age</label>
          <input type="number" id="age" name="age" placeholder="e.g. 20" min="10" max="100" />
          <span class="field-error-text" data-error-for="age"></span>
        </div>
      </div>
    </form>
  </div>
</main>
```

```
</div>
```

```
<div class="field">
```

```
<label for="dob">Date of Birth</label>
```

```
<input type="date" id="dob" name="dob" />
```

```
<span class="field-error-text" data-error-for="dob"></span>
```

```
</div>
```

```
<div class="field full">
```

```
<label for="gender">Gender</label>
```

```
<select id="gender" name="gender">
```

```
<option value="">Select gender</option>
```

```
<option value="Male">Male</option>
```

```
<option value="Female">Female</option>
```

```
<option value="Other">Other</option>
```

```
<option value="Prefer not to say">Prefer not to say</option>
```

```
</select>
```

```
<span class="field-error-text" data-error-for="gender"></span>
```

```
</div>
```

```
</div>
```

```
<div class="form-section-title">Contact Information</div>
```

```
<div class="form-grid">
```

```
<div class="field">
```

```
<label for="email">Email</label>
```

```
<input type="email" id="email" name="email" placeholder="e.g. rahul@gmail.com" />
```

```
<span class="field-error-text" data-error-for="email"></span>
```

```
</div>
```

```
<div class="field">
```

```
<label for="mobile">Mobile Number</label>
```

```
<input type="tel" id="mobile" name="mobile" placeholder="e.g. 9876543210" />
```

```
<span class="field-error-text" data-error-for="mobile"></span>
```

```
</div>
```

```
<div class="field full">
```

```
<label for="address">Address</label>
```

```
<textarea id="address" name="address" placeholder="City, State"></textarea>
```

```
<span class="field-error-text" data-error-for="address"></span>
```

```
</div>
```

```
</div>
```

```
<div class="form-section-title">Academic Information</div>
```

```
<div class="form-grid">
```

```
<div class="field">
```

```
<label for="course">Course</label>
```

```
<input type="text" id="course" name="course" placeholder="e.g. B.Tech CSE" />
```

```
<span class="field-error-text" data-error-for="course"></span>
```

```
</div>
```

```
<div class="field">
```

```
<label for="year">Year of Study</label>
```

```
<select id="year" name="year">
```

```
<option value="">Select year</option>
```

```
<option value="1st Year">1st Year</option>
```

```
<option value="2nd Year">2nd Year</option>
```

```
<option value="3rd Year">3rd Year</option>
```

```
<option value="4th Year">4th Year</option>
```

```
</select>
```

```
<span class="field-error-text" data-error-for="year"></span>
```

```
</div>
```

```
</div>
```

```
<div class="form-section-title">Account Information</div>
```

```
<div class="form-grid">
```

```
<div class="field">
```

```
<label for="username">Username</label>
```

```
<input type="text" id="username" name="username" placeholder="e.g. rahul123" />
```

```
<span class="field-error-text" data-error-for="username"></span>
```

```
</div>
```

```

<div class="field">
  <label for="password">Password</label>
  <input type="password" id="password" name="password" placeholder="At least 6 characters" />
  <small class="hint">Minimum 6 characters.</small>
  <span class="field-error-text" data-error-for="password"></span>
</div>
</div>
<button type="submit" class="btn btn-primary btn-block" id="registerBtn">Create Account</button>
<div class="form-footer-link">
  Already have an account? <a href="login.html">Login</a>
</div>
</form>
</div>
</main>
<script src="js/script.js"></script>
</body>
</html>

```

script.js

```

function showMessage(element, text, type) {
  element.textContent = text;
  element.className = "message-box show " + type;
}

/** Clears any message currently shown. */
function hideMessage(element) {
  element.className = "message-box";
  element.textContent = "";
}

/** Displays an inline error under a specific field, and highlights it. */
function setFieldError(fieldName, message) {
  const errorEl = document.querySelector(`[data-error-for="${fieldName}"]`);
  const inputEl = document.getElementById(fieldName);

```

```

if (errorEl) errorEl.textContent = message || "";

if (inputEl) {
    const wrapper = inputEl.closest(".field");
    if (wrapper) wrapper.classList.toggle("has-error", Boolean(message));
}
}

/** Clears all inline field errors within a form. */
function clearFieldErrors(form) {
    form.querySelectorAll(".field-error-text").forEach((el) => (el.textContent = ""));
    form.querySelectorAll(".field.has-error").forEach((el) => el.classList.remove("has-error"));
}

const EMAIL_REGEX = /^[^\s@]+@[^\s@]+\.[^\s@]+$/;

// =====
// REGISTRATION PAGE LOGIC
// =====

const registerForm = document.getElementById("registerForm");
if (registerForm) {
    const formMessage = document.getElementById("formMessage");
    const registerBtn = document.getElementById("registerBtn");
    registerForm.addEventListener("submit", async (event) => {
        event.preventDefault();
        hideMessage(formMessage);
        clearFieldErrors(registerForm);
        // ---- Gather field values ----
        const data = {
            name: document.getElementById("name").value.trim(),
            age: document.getElementById("age").value.trim(),
            dob: document.getElementById("dob").value,
            gender: document.getElementById("gender").value,
            email: document.getElementById("email").value.trim(),
            mobile: document.getElementById("mobile").value.trim(),

```

```
address: document.getElementById("address").value.trim(),
course: document.getElementById("course").value.trim(),
year: document.getElementById("year").value,
username: document.getElementById("username").value.trim(),
password: document.getElementById("password").value,
};

// ---- Frontend validation ----

let isValid = true;

const requiredFields = [
  "name", "age", "dob", "gender", "email",
  "mobile", "address", "course", "year", "username", "password",
];

requiredFields.forEach((field) => {
  if (!data[field]) {
    setFieldError(field, "This field is required.");
    isValid = false;
  }
});

if (data.age && (Number(data.age) < 10 || Number(data.age) > 100)) {
  setFieldError("age", "Enter a realistic age.");
  isValid = false;
}

if (data.email && !EMAIL_REGEX.test(data.email)) {
  setFieldError("email", "Enter a valid email address.");
  isValid = false;
}

const mobileDigits = data.mobile.replace(/\D/g, "");

if (data.mobile && (mobileDigits.length < 7 || mobileDigits.length > 15)) {
  setFieldError("mobile", "Enter a valid mobile number.");
  isValid = false;
}
```

```
if (data.password && data.password.length < 6) {
  setFieldError("password", "Password must be at least 6 characters.");
  isValid = false;
}
if (!isValid) {
  showMessage(formMessage, "Please fix the highlighted fields.", "error");
  return;
}
// ---- Send to backend ----
registerBtn.disabled = true;
registerBtn.textContent = "Creating account...";
try {
  const response = await fetch("/register", {
    method: "POST",
    headers: { "Content-Type": "application/json" },
    body: JSON.stringify(data),
  });
  const result = await response.json();
  if (result.success) {
    showMessage(formMessage, "Registration successful! Redirecting to login...", "success");
    registerForm.reset();
    setTimeout(() => {
      window.location.href = "/login";
    }, 1200);
  } else {
    showMessage(formMessage, result.message || "Registration failed.", "error");
  }
} catch (err) {
  console.error("Registration request failed:", err);
  showMessage(formMessage, "Could not reach the server. Please try again.", "error");
} finally {
```

```
    registerBtn.disabled = false;

    registerBtn.textContent = "Create Account";

  }

});

}

// =====
// LOGIN PAGE LOGIC
// =====

const loginForm = document.getElementById("loginForm");
if (loginForm) {
  const formMessage = document.getElementById("formMessage");
  const loginBtn = document.getElementById("loginBtn");
  loginForm.addEventListener("submit", async (event) => {
    event.preventDefault();
    hideMessage(formMessage);
    clearFieldErrors(loginForm);
    const identifier = document.getElementById("identifier").value.trim();
    const password = document.getElementById("password").value;
    let isValid = true;
    if (!identifier) {
      setFieldError("identifier", "Enter your username or email.");
      isValid = false;
    }
    if (!password) {
      setFieldError("password", "Enter your password.");
      isValid = false;
    }
    if (!isValid) {
      showMessage(formMessage, "Please fill in both fields.", "error");
      return;
    }
  })
}
```

```
loginBtn.disabled = true;

loginBtn.textContent = "Logging in...";

try {
  const response = await fetch("/login", {
    method: "POST",
    headers: { "Content-Type": "application/json" },
    body: JSON.stringify({ identifier, password }),
  });

  const result = await response.json();

  if (result.success) {
    // Store the logged-in user's info for the dashboard to read
    localStorage.setItem("loggedInUser", JSON.stringify(result.user));
    showMessage(formMessage, "Login successful! Redirecting...", "success");
    setTimeout(() => {
      window.location.href = "/dashboard.html";
    }, 600);
  } else {
    showMessage(formMessage, result.message || "Invalid username/email or password", "error");
  }
} catch (err) {
  console.error("Login request failed:", err);
  showMessage(formMessage, "Could not reach the server. Please try again.", "error");
} finally {
  loginBtn.disabled = false;
  loginBtn.textContent = "Login";
}

});

// =====
// DASHBOARD PAGE LOGIC
// =====
```

```
const dashboardContent = document.getElementById("dashboardContent");
if (dashboardContent) {
  const loadingState = document.getElementById("loadingState");
  const logoutBtn = document.getElementById("logoutBtn");
  function formatDob(dobString) {
    if (!dobString) return "—";
    const date = new Date(dobString);
    if (Number.isNaN(date.getTime())) return dobString;
    return date.toLocaleDateString("en-GB", { day: "numeric", month: "long", year: "numeric" });
  }
  function loadDashboard() {
    const stored = localStorage.getItem("loggedInUser");
    // No logged-in user found -> send back to login
    if (!stored) {
      window.location.href = "/login";
      return;
    }
    let user;
    try {
      user = JSON.parse(stored);
    } catch (err) {
      console.error("Corrupted session data:", err);
      localStorage.removeItem("loggedInUser");
      window.location.href = "/login";
      return;
    }
    if (!user || !user.username) {
      window.location.href = "/login";
      return;
    }
    // Populate the dashboard with this user's own information only
```

```
document.getElementById("welcomeHeading").textContent = `Welcome, ${user.name}!`;
document.getElementById("avatarInitial").textContent = user.name.charAt(0).toUpperCase();
document.getElementById("profileName").textContent = user.name;
document.getElementById("profileCourseYear").textContent = `${user.course} • ${user.year}`;
document.getElementById("fieldName").textContent = user.name || "—";
document.getElementById("fieldEmail").textContent = user.email || "—";
document.getElementById("fieldUsername").textContent = user.username || "—";
document.getElementById("fieldMobile").textContent = user.mobile || "—";
document.getElementById("fieldDob").textContent = formatDob(user.dob);
document.getElementById("fieldGender").textContent = user.gender || "—";
document.getElementById("fieldCourse").textContent = user.course || "—";
document.getElementById("fieldYear").textContent = user.year || "—";
document.getElementById("fieldAddress").textContent = user.address || "—";

loadingState.style.display = "none";
dashboardContent.style.display = "block";
}
logoutBtn.addEventListener("click", () => {
  localStorage.removeItem("loggedInUser");
  window.location.href = "/login";
});
loadDashboard();
}
```

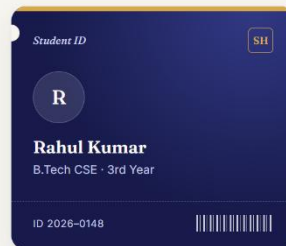
Output

Manage Your Student Journey

StudentHub keeps your academic profile in one place — register once, log in anytime, and view your personal and academic information from a clean, simple dashboard.

[Get Started](#)

[Login](#)



Three steps, start to finish

From first registration to your personal dashboard, the whole journey happens in one place.

1

Easy Registration

Quickly create a student profile with your personal, contact, and academic details.

2

Secure Login

Access the student dashboard using your username or email along with your password.

3

Student Dashboard

View your personal and academic information in one clean, organized profile card.

SH

Welcome back to your record

Log in to pick up right where you left off — your profile, course, and details, exactly as you registered them.

“One login, your whole student profile.”

Login

Enter your credentials to access your dashboard.

Username or Email

e.g. rahul123 or rahul@gmail.com

Password

Your password

[Login](#)

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Contact Information

Email

e.g. rahul@gmail.com

Mobile Number

e.g. 9876543210

Address

City, State

Academic Information

Course

e.g. B.Tech CSE

Year of Study

Select year

Account Information

Username

e.g. rahul123

Password

At least 6 characters

Minimum 6 characters.

Create Account

Already have an account? [Login](#)